
VISUAL HALLUCINATION IN LARGE VISION-LANGUAGE MODELS

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1 Introduction

Large Vision-Language Models (LVLMs), such as GPT-4 vision [1] and Llava [2], have made substantial advances in synchronizing visual and textual features. These models exhibit impressive abilities of understanding images and communicating in natural languages. However, they encounter challenges with visual hallucinations in their generation. Visual hallucination is featured by generating fictitious information, such as non-existent objects or incorrect categories, attributes, or relationships. This problem can negatively impact many vision-language tasks, such as visual reasoning, and degrade applications that require highly accurate image descriptions, including robotics and medical imaging.

We categorize visual hallucination into two types: content-based hallucination and knowledge-based hallucination, inspired by a study of large language models [3]. For instance, the answer to questions like “Is there a cat in the image?” is largely deterministic, and so uncertainty on this prompt can be called content-based, or epistemic. If visual hallucination occurs within such a question (intuitively, the question can be answered by simply looking at the images) can be reduced by enhancing the capabilities of the models themselves. In many other cases, however, the probability distribution of the generated answer exhibits inherent entropy. For example, a question that has a seemingly tangent connection with the given image, e.g., the question asking “When is the time for building this bridge?” for an image. Such questions are not obvious from looking at the images, and the resulting hallucination is termed knowledge-based hallucination. Unlike content-based hallucination, this knowledge-based uncertainty is not reducible through simply training the model: it does not disappear even in the limit of infinitely large models that have been trained with infinite data [3]. However, this knowledge-based hallucination can be mitigated through training a better model that have better awareness of its self-knowledge [4], and retrieving external sources for answers.

Previous research works have investigated visual hallucination in various aspects, including evaluation and benchmarking [5, 6], mitigating content-based hallucination [7, 8, 9], and addressing knowledge-based visual hallucination through retrieval augmented generation [10]. However, they fall short of comprehensively understanding the issue of visual hallucination. Such lack of understanding is two-fold: 1) the difference between content-based hallucination and knowledge-based hallucination; and 2) the identification of causes of visual hallucination.

2 Research Questions & Directions

To facilitate understanding of visual hallucination in LVLMs, we guide this study with the following questions:

Analysis of content-based hallucination and knowledge-based hallucination. We aim to have a better understanding and characterization of these two types of hallucination. Our research goals include:

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1. Identifying the characteristics of these two types of hallucination. For example, we expect output from knowledge-based hallucinations to have larger entropy compared to content-based hallucinations.
2. Establishing a formal definition and measurement of these two types of hallucination.

Causes of visual hallucination. This question involves the identification of the causes of visual hallucination. Specifically, we suggest to investigate how each component of LVLMs contributes to this problem. The mainstream open-source LVLMs, such as Llava [2] and Mini-GPT4 [11], consist of three main components:

1. *Vision Encoder*: This component processes the image and is usually pre-trained in image tasks. Its failure may result in the ignorance of fine-grained details in images, leading to content-based hallucinations.
2. *Mapping Module*: This component bridges the vision encoder to the language model and is typically an MLP. A failure here may result in a mismatch between the text and image modalities.
3. *Language Model*: This LLM component processes both language tokens and output from the mapping module. Failure in this component can lead to the generation of spurious content that is only loosely correlated with the input, resulting in knowledge-based hallucinations.

We intend to isolate and identify the hallucination from these different failure modes.

Experiments. The initial sets of experiments include:

1. Finding or constructing datasets that lead to the two different types of hallucination; measuring the statistical property and characteristics of the two types of hallucination.
2. Isolating effects from the three components of LVLMs. For example, varying the vision encoders (e.g., CLIP [12] and DINO [13]) or fine-tuning the language model via supervised learning or KTO [14] to see its effects on hallucination performance.
3. Optional - mechanistically understanding the internal circuits of LVLMs that lead to different hallucinations, e.g. putting a linear probe in different hidden layers to detect different hallucinations.
4. Optional - proposing methods to mitigate hallucination after understanding its cause.

3 Outcomes

In this project, we investigate the nuances of visual hallucination, providing a comprehensive analysis that illuminates this issue’s intricacies. Our investigation first offers a detailed understanding of both content-based and knowledge-based hallucinations, elaborating on the underlying statistical properties of each kind. We further delve into components of LVLMs, exploring how each component could potentially contribute to visual hallucination. A possible extension of our study can be a comparative study of approaches to mitigating visual hallucinations, which outlines the advantages and limitations of each method and provides insights into how these approaches effectively reduce content-based uncertainty. We intend to contribute significant insights into understanding visual hallucinations.

4 Pre-requisite

Given the scope of the projects, it would be beneficial for the students to have prior experience in LVLMs such as Blip-2 [15], Llava [2], and Mini-GPT4 [11]. Since LVLMs are visual-conditioned language models, prior experience with large language models will be highly desirable. In addition, since this project requires understanding the circuits of neural networks, experience with mechanistic interpretability would be important. Students are *strongly* advised to consult the project’s advisor beforehand.

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